Physical Science Part 2

1. A week attraction between hydrogen atom in one molecule and an oxygen in other called________.
A. covalent bond
B. Hydrogen bond
C. Ionic bond
D. Metallic bond

2. Which of the following substances is most likely a covalent compound ?
A. $CaCO_3$
B. KNO_3
C. SO_2
D. NaOH

3. What is the element needed to bond with Na to create the most ionic bond?
A. F
B. C1
C. Br
D. I

4. The boiling point of substance A is higher than that of substance B. What does this meaning?
A. Substance A is less viscous than substance B.
B. Substance A evaporates faster than substance B.
C. Substance B has greater surface tension than substance A.
D. Substance B has weaker intermolecular forces than substance A.

5. What is the intermolecular force of attraction between the nucleus of one atom and the electrons of a nearby atom?
A. Dipole-dipole force
B. Ion-dipole force
C. London dispersion force
D. Van der Waals force

6. Which of the following has the greatest number of atom?
A. $Ca(OH)_2$
B. $A1_2O_3$
C. $NaHCO_3$
D. C_2H_2

7. What does a triangle , which is placed in a chemical equation indicate?
A. Changes happened
B. Heat is applied
C. Gas is evolved
D. Precipitate

8. What would most probably be the resulting chemical formula when $Mg^{+2}$and O^{-2}combine?
A. MgO
B. Mg_2O_2
C. MgO_2
D. Mg_4O_4

9. Which of the following is an example of combustion reaction?
A. $Mg+C1_2$ ------------------------------------> $MgC1_2$
B. $CaCO_3$ ------------------------------------> $CaO + CO_2$
C. $2\ C_4H_{10} + 13\ O_2$ ---------------------------------> $8\ CO_2 + 10\ H_2O$
D. $AgNO_3$ + NaC1 -------------------------------> AgC1 + $NaNO_3$

10. What type of chemical reaction is involved in $Ca_{(S)} + 2H_2O\ Ca(OH)_{2(aq)}$?
A. Combination
B. Decomposition
C. Single-displacement
D. Double-displacement

11. When the fuel in a butane gas stove undergoes complete combustion, it reacts with oxygen to form carbon dioxide and water:_C_4H_{10} + _ O_2CO_2+_H_2O. Which set of coefficients can be BEST used in balancing the equation?
A. 1, 1, 4, 2
B. 2, 2, 4, 5
C. 2, 8, 13, 10
D. 2, 13, 8, 10

12. In oxidation-reduction, the oxidizing agents is reduced and gains electrons. The reducing agent is oxidized and loses electrons. Are the statements CORRECT?
A. Both statements are correct.
B. Both statements are false
C. First statement is true; second statement is false.
D. Second statement is true; first statement is false.

13. How many moles of iron does 25 g of Fe represent? (Fe=55.85 g)
A. 0.448 mol
B. 2.234 mol
C. 30.85 mol
D. 80.85 mol
Ans: A. 0.448 mol
- (25 g) / (55.85 g/mol) =0.448 mol

14. What is the mass of 5.0 mol of water? (1 mol H_2O=18.02g)
A. 3.60 g

B. 13.02 g
C.23.02 g
D. 90.10 g
Ans: C. 90.10 g
-(5 mol) (18.02 g/mol) = 90.10 g

15. An analysis of salt shows that it contains 56.58 g potassium; 8.68 g carbon and 34.73 oxygen. What is the empirical formula of the substance?
A. KC_2O_3
B. K_2CO_2
C. K_2CO_3
D. K_2C_3O

16. Which of the following is NOT a principal assumption of Kinetic Molecular Theory?
A. Gas particles move rapidly.
B. Gases consists of submicroscopic particles.
C. There is very little empty space in a gas.
D. Gas particles have no attraction for one another.

17. Based on the Kinetic Molecular Theory, a gas can be compressed much more than a solid or liquid because___________.
A. gas particles move rapidly
B. the particles of a gas are very far apart
C. a gas is composed do not attract or repel one another
D. gas particles do not attract or repel one another

18. Which refers to the kinetic energy or motion of gas particles?
A. Pressure
B. Quantity of gas
C. Temperature
D. Volume

19. A given mass of gas has a volume of 750 ml at -25℃ and 280 torr. What would the volume of the gas be at 30 and 560 torr of pressure considering the amount of gas which is constant?
A. 450 ml
B. 458 ml
C. 480 ml
D. 520 ml
Ans: B. 458 ml
This item involves combined gas law : $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

20. Seven liters of an ideal gas are contained at 3.5 atm and 30 How many moles of this gas are present?
A. 0.50 mol

B. 0.75 mol
C. 1.00 mol
D. 1. 25 mol

21. What is the BEST reason why the atmospheric pressure in Baguio City is lower than the atmospheric pressure at sea level?
A. Because there are more gas particles at higher altitudes
B. Because there are less gas particles at higher altitudes
C. Because there are greater temperature at higher altitudes
D. Because there is lesser temperature at higher altitudes

22. Which of the following correctly describes the process of exhalation (air going out of the lungs)?
A. The lungs decrease in volume, causing the internal pressure to increase.
B. The lungs decrease in volume, causing the internal pressure to decrease.
C. The lungs increase in volume, causing the internal pressure to increase.
D. The lungs increase in volume, causing the internal pressure to decrease.

23. What is the closed, continuous path through which electrons can flow?

A. Charge
B. Circuit
C. Resistor
D. Voltage

24. What will be your charge if you scrape electrons from your feet while scuffing across the rug?
A. Negative
B. Neutral
C. Positive
D. Not enough information given

25. How is food cooked faster using a pressure cooker?
A. The accumulated steam increases the pressure and decreases the boiling point of the liquid.
B. The accumulated steam increases the pressure and increases the boiling point of the liquid.
C. The accumulated steam decreases the pressure and increases the boiling point of the liquid.
D. The accumulated steam decreases the pressure and decreases the boiling point of the liquid.

26. Which of the following is NOT a characteristic of a base?
A. Bitter taste
B. Slippery feeling
C. Able to change litmus from red to blue

D. Reacts with carbonates to produce carbon dioxide

27. Which of the following statements about acid and base is TRUE?
A. Baking soda is acidic.
B. Both acids and bases conduct electricity.
C. The concept of H_3O^+ was introduced by Arrhenius.
D. The end point is reach when the color of the indicator in titration changes.

28. What is the pH of the solution with an $[H^+]$ of 1.00×10^{-11}?
A. 1
B. 3
C. 7
D. 11

29. Which of the following is the most acidic?
A. Calamansi juice
B. Carbonated drink
C. Urine
D. Vinegar

30. Acid rain is formed when sulfur oxides react with water in air. It causes lakes and streams decrease in pH which makes it unable to support fish and plant life. Which is the LEAST means of solving this problem?
A. Limestone may be added to the water basins.
B. Neutralize sulfur oxides with basic compounds.
C. Reduction of sulfur emissions from the industrial establishments.
D. Providing absorbent colloidal materials to screen acid rain before hitting the water.

31. Which of the following would decrease the rate in which a solid dissolves in a liquid?
A. Supersaturating the solution
B. Applying pressure into the solution
C. Grinding the solid into smaller pieces
D. Increasing the temperature of an exothermic reaction

32. Which factors affects the rate of dissolution when powdered sugar dissolves more rapidly than granulated sugar?
A. Application of heat
B. Nature of solute
C. Particle size
D. Stirring

33. In which factor is the description “like dissolves like” BEST used?

A. Volume
B. Pressure
C. Temperature
D. Nature of solute and solvent

34. When the production of hydrogen ions in the body is the same as their loss, you are in__________.
A. acid-base balance
B. electricity balance
C. water balance
D. balance diet

35.What is the mass percent of Na_2SO_4 in a solution made by dissolving 25 g Na_2SO_4 in 225 g H_2O?
A. .01%
B. 1%
C. 10%
D.20%

36. What is the mole fraction of 100 g ethanol (C_2H_6O) mixed in 100 g water?(C=12 H=1 O=16)
A. 0.28
B. 2.17
C. 5.56
D.7.73

37. What is the molarity of the solution containing 1.4 mol of acetic acid ($HC_2H_3O_2$) in 250 ml of solution?
A. 4.5M
B. 5.6M
C. 6.5M
D. 7.6M

38. What is the normality of a 2M solution of H_3PO_4?
A. 2 N
B. 3 N
C. 5 N
D. 6 N

39. When egg yolk is added to oil and water vinegar to make mayonnaise, the egg yolk serves as___________.
A. coagulant
B. emulsifying agent
C. solvent
D. surfactant

40. Which of the following involves adsorption?
A. Plating the copper on a steel object
B. Adherence of paint to wood surfaces
C. Brown color of the eyes of the Filipinos
D. Remobal of odor inside a refrigerator using charcoal

Answer Keys:

1. B
2. C
3. A
4. D
5. C
6. C
7. B
8. A
9. C
10. C
11. D
12. A
13. A
14. C
15. C
16. C
17. B
18. C
19. B
20. C
21. B
22. A
23. B
24. C
25. B
26. D
27. B
28. D
29. A
30. D
31. D
32. C
33. D
34. B
35. C
36. A
37. B
38. D
39. B
40. D